Water Use and Technical Efficiencies in Private Irrigated Perimeters in *Zeuss-Koutine Watershed,* South-Eastern Tunisia

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Abstract

Overpumping of the Smar and Jorf Aquifers in Zeuss-Koutine region due expansion of irrigated agriculture is believed to have resulted in reductions in water shortage volumes and drops in water table in several regions. It is widely agreed upon that water demand should be reduced through increases in water use efficiency to make irrigated agriculture in the region more sustainable. For that, many reforms have been made in Tunisia in the last two decades to improve irrigation water efficiency. The purpose of this paper is to asses the technical efficiency (TE) and proposes a measure for irrigation water efficiency (IE) based on the concept of input-specific technical efficiency for a sample of 100 irrigators in Zeuss-Koutine region (south-eastern of Tunisia). The method used to quantify TE and IE, is Data Envelopment Analysis (DEA). Based upon linear programming techniques, DEA creates a "best practice" production frontier based on the irrigators that produced their level of crop output with the least amount of (all) farm inputs used (TE) and with least amount of water when other inputs are constant(IE).

A major finding of the study is that the irrigation systems are clearly inefficient. Under the CRS assumption, the average technical efficiency of the sample was 64%. This implies that output level could be producing by saving 36% of (all) farm inputs. The calculated irrigation water use is still low (47, 8%) and does not reflect the water-saving oriented policies that have been applied. Considering that the mean irrigation water use per farm was 4652.90 m^3/ha in the study area, this infers that almost half of water used was "excess".

Keywords: irrigation, technical efficiency, DEA method, irrigation water efficiency, Aquifers, Tunisia

Introduction

Irrigation water is becoming an increasingly scarce resource for agriculture in many regions of the world. Common to past policy schemes was the development of an adequate irrigation infrastructure to guarantee the supply of irrigation water as the demand for agricultural products increased. However, these expansionary policies have led to a massive use of irrigation water at a heavily subsidized cost, and a scarcity of the resource. Water shortage has become an increasing social and economic concern for policy makers and for those who must compete for the resource. In particular, policy makers are beginning to point to agriculture as the sector at the core of the water problem. Tunisian water reserves are estimated at 4.7 billion m^3/year, of which 2.7 billion m^3 comes from annual rivers in the north, 0.7 billion m^3 from groundwater in the centre, the plains and the coastal area, and approximately 1.3 billion m^3 from the deep groundwater table mainly in the south. Water resources are unevenly distributed across the country, with around 60% located in the north, 18% in the centre and 22% in the south. Water resources that have a salinity of less than 1.5 g/liter are distributed as follows: 72% of surface water resources, 8% of shallow groundwater and 20% of deep groundwater.

Taking into account the limited water resources and the frequent disparity between supply and demand during dry seasons, Tunisia has engaged over the last three decades in a dynamic program of water mobilization. Several investment projects have been granted, reaching 9% of total investments in the government's Development Plan X (2002-2006), in which it has invested 19% in water programs). Agriculture, which accounts for approximately 12% of the GDP, is the sector that consumes the most water (more than 80% of the total demand).

Currently, almost 385,000 ha (7% of useful agricultural land) are irrigated in Tunisia. The irrigation sector consumes 80% of the available water resources and provides 35% of the agricultural production value, 95% of horticultural crops, 30% of dairy production, almost 22% of agricultural sector exports, and 26% of total agricultural employment (Ministry of Agriculture and Water Resources, 2004). Moreover, the demand for irrigation water continues to rise, due largely to the development of new irrigation projects and the intensification of irrigation within existing areas. Therefore, during recent decades concerns regarding the efficient use of water resources in the country have increased.

These concerns have been addressed in three ways: (i) modernizing the management of collective irrigation systems by enhancing the role played by water users' associations and by promoting user participation in all aspects of management, (ii) reformulating the water pricing system by introducing a cost recovery objective and (iii) developing incentives to enhance and promote the adoption of water saving technologies at farm level (Al Atiri, 2004).
In this context, the objective of this paper is to estimate and assess the technical efficiency (TE) and irrigation water efficiency (IE) based on the concept of input-specific technical efficiency of a sample of 100 private irrigators in Zeuss-Koutine region (south-eastern of Tunisia) using Data Envelopment Analysis (DEA). Based upon linear programming techniques, DEA creates a "best practice" production frontier based on the irrigators that produced their level of crop output with the least amount of (all) farm inputs used (TE) and with least amount of water when other inputs are constant(IE). What is implied is that those who are able to produce their output levels using, the least amount of (all) farms inputs or least amount of water when other inputs are kept constant, are efficient and better water manager.
This choice of the study area is motivated by the expansion of irrigated agriculture, the predominance of private initiative in the development of irrigation (Bruno et al, 2004) and by the overpumping of the Smar and Jorf Aquifers in Zeuss-Koutine region (DRE, 2005).

2. Methodology

2.1 Efficiency measures

The absolute efficiency position of farmers is usually not known. Therefore the problem is to measure the efficiency of one farm relative to others. The measurement of technical efficiency is based upon deviations of observed output or input vectors from the best production or efficient production frontier. If a production units' actual production point lies on the frontier it is perfectly efficient. If it deviates from the frontier then it is technically inefficient, with the ratio of the actual to potential production defining the level of efficiency of the individual firm. Our measure of technical efficiency provides an indication of how the use of all inputs can be minimized in the production process of a given farm, while continuing to produce the same level of output.
Additionally, we consider the possible reduction of a subset of inputs while keeping other inputs and the output constant. This generates a ''sub-vector efficiency'' measure which only estimates the relative input reduction potentials in a subset of the inputs, in this case irrigation water alone, rather than the reduction potential in all inputs simultaneously. The efficiency measure produced can be called ''water use efficiency'' or in the case of irrigated, production, ''irrigation water efficiency'' (IE).

There are two main competing paradigms for estimating the relative efficiency of farms: parametric and non-parametric. The results from both methods are highly correlated in most cases (Wadud and White, 2000; Alene and Zeller, 2005), indicating that both methods are valuable and the choice can be based on a researcher's preference. A major advantage of non-parametric DEA for this study is that the calculation of sub-vector efficiency for irrigation water use is relatively straightforward (Speelman et al., 2008).

2.2 Data Envelopment Analysis

The evaluation of farm (the decision-making unit) performance is usually based on economic efficiency, which is generally composed of two major components: technical efficiency and price or allocative efficiency. Technical efficiency is defined as the ability of a farm to either produce the maximum possible output from a given bundle of inputs and a given technology, or to produce the given level of output from the minimum amount of inputs for a given technology. Technical efficiency can be decomposed into two components: pure technical efficiency and scale efficiency. When one separates the scale effect from the technical efficiency, the pure technical efficiency is obtained. Scale efficiency relates to the most efficient scale of operation in the sense of maximizing average productivity. A scale efficient farm has the same level of technical and pure technical efficiency. Allocative efficiency is defined as the ability of a farm to equate marginal value product and marginal cost.
In the present paper, we focus on technical efficiency measure with input-oriented DEA models because, in the context of increasing water scarcity, it is more relevant to consider potential decreases in water use than increases in output (Frija et al, 2009).

Following the Banker et al. (1984) BCC–DEA model, and considering a data set of N farms (j= 1…, n), each of them using M inputs x (m = 1, . . ., M), for producing S outputs y (j = 1, . . .,S), each farm becomes the reference unit. We then solve the following linear program (1) J times (once for each

farm). The model specification (1) is the dual form of an equivalent primal model specification that maximizes the outputs for given inputs.

$$
\begin{aligned}
&\mathrm{Min}_{\theta,\lambda}\theta \\
&\text{s.t} \\
&\sum_{j=1}^{n} \lambda^{j} y_{s}^{j} \geq y_{s}^{1} \\
&\sum_{j=1}^{n} \lambda^{j} x_{m}^{j} \leq \theta . x_{m}^{1} \qquad (1) \\
&\sum_{i=1}^{n} \lambda^{j} = 1 \\
&\lambda^{j} \geq 0
\end{aligned}
$$

Where θ represents technical efficiency and hence the percentage of radial reduction to which each of the inputs is subjected; λ_j is a vector of j elements representing the influence of each farm in determining the technical efficiency of the farm under study ($farm_0$), x_{n1} and y_{m1} are, respectively, the input and the output vectors of the $farm_1$. The equation $\sum_{i}^{J} \lambda_j = 1$ is a convexity constraint, which specifies the VRS framework (pure technical efficiency). Without this convexity constraint, the DEA model will be a CCR model describing a CRS situation (The overall technical efficiency). In addition, a comparison of both scores is interesting because it provides information on scale efficiency (SE). Coelli et al. (2002) showed that the relation is as follows: $SE = \frac{\theta^{CRS}}{\theta^{VRS}}$

The notion of sub-vector efficiency is introduced to account for a specific IE score for each farm (Speelman et al., 2008; Lilienfeld and Asmild, 2007; Oude Lansink and Silva, 2004; Oude Lansink et al., 2002). It's only estimates the relative input reduction potentials in a subset of the inputs, in this case irrigation water alone, rather than the reduction potential in all inputs simultaneously. With irrigation water as the first input, the irrigation water-specific DEA efficiency score for observation ^{k}, θ^{k} is calculated by the following linear programming (LP) problem:

$$
\begin{aligned}
&\theta^{k} = \min_{\theta,\lambda} \theta \\
&\text{s.t.} \\
&\theta x_{1}^{k} - \sum_{j=1}^{n} \lambda^{j} x_{1}^{j} \geq 0 \\
&x_{m}^{k} - \sum_{j=1}^{n} \lambda^{j} x_{m}^{j} \geq 0, \ \ \mathrm{m=1.....M} \qquad (2) \\
&-y_{s}^{k} + \sum_{j=1}^{n} \lambda^{j} y_{s}^{j} \geq 0, \ \ \mathrm{s=1.....S} \\
&\sum_{j=1}^{n} \lambda_{j} = 1 \\
&\lambda^{j} \geq 0
\end{aligned}
$$

A DEA efficiency score, given by a specific θ, can have a value between 0 and 1 where a value of 1 indicates that the observation is a best performer located on the production frontier and has no reduction potential on irrigation water. Any value of θ smaller than 1, however, indicates water use inefficiency, i.e., that excess irrigation water is being used.

Based upon linear programming techniques, DEA creates a “best practice” production frontier based on the irrigators that produced their level of crop output with the least amount of water. What is implied is that those who are able to produce their output levels using the least amount of water are better water manager. These farms then serve as benchmarks against witch the water use inefficiency of all other irrigators, or amount of “excess water” used, can be measured. As an example, a θ value of 0.8 means that the observation should be able to produce the same level of output using 80% of its current level of irrigation water when compared to its benchmark which is constructed from the best performers with similar characteristics. The excess water used can then be calculated as $(1-\theta)x_1$ which in the previous example means that the excess is 20% of the current amount of irrigation water used.

The difference between the technical and the sub-vector efficiency in DEA approach is illustrated in Fig. 2. We consider three farms using two inputs (water and another input) and producing a single output. Farms B1 and B2 are the best performers and linear combinations of their input use define the production frontier which envelopes all the observed data. The observation A is, however, inefficient. The radial contraction of the input vector xi [water, input2] produces a projected point on the frontier (A0). The obtained point A0 is a linear combination of the observed data points B1 and B2, with the constraints (7, 8 and 9) in program 1, ensuring that the projected point cannot lie outside the feasible set. The technical efficiency of farm A, relative to B1 and B2, is given by the ratio $\theta^t = OA^0/OA$.

The irrigation water use inefficiency (IE) of farm A could be demonstrated using a scenario in which water quantity is reduced while keeping input 2 and the output constant. In this case A is projected to A’ and the IE is given by the ratio θ^t = O’A’/O’A. The horizontal distance to the frontier represents their excess use of the water input compared to their benchmarks on the frontier which produce the same output using less water.
The benchmark is reached by multiplying an observation’s water input by its estimated DEA efficiency score θ. If observation A, for example, uses 5 units of water to produce its current level of output, then an estimated θ value of 0.6 would indicate that the benchmark (here a convex combination of observations B_1 and B_2) uses 0.6 x 5 = 3 units of water to produce the same amount of output. The distance between the observation A and its benchmark A’ indicates the excess water used, which would here be 5-3 = 2 units of water, as compared to a benchmark that produces the same level of outputs. This also implies that A can save 2 units of irrigation water.

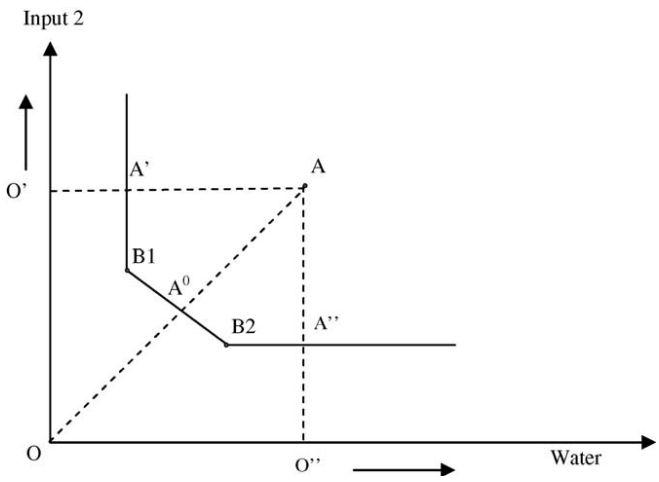

Figure 1. Differences between technical efficiency and irrigation water efficiency input-oriented efficiency (based on Oude Lansink et al., 2002).

2.3 Case study and data collection

We collected our data from small-scale private irrigated farms in the region of Zeuss-Koutine, located in south-eastern area of Tunisia and within governorate of Médenine (cf.fig.2). In this region irrigation activity is recently introduced and water scarcity is an important issue (Bruno et al, 2004).

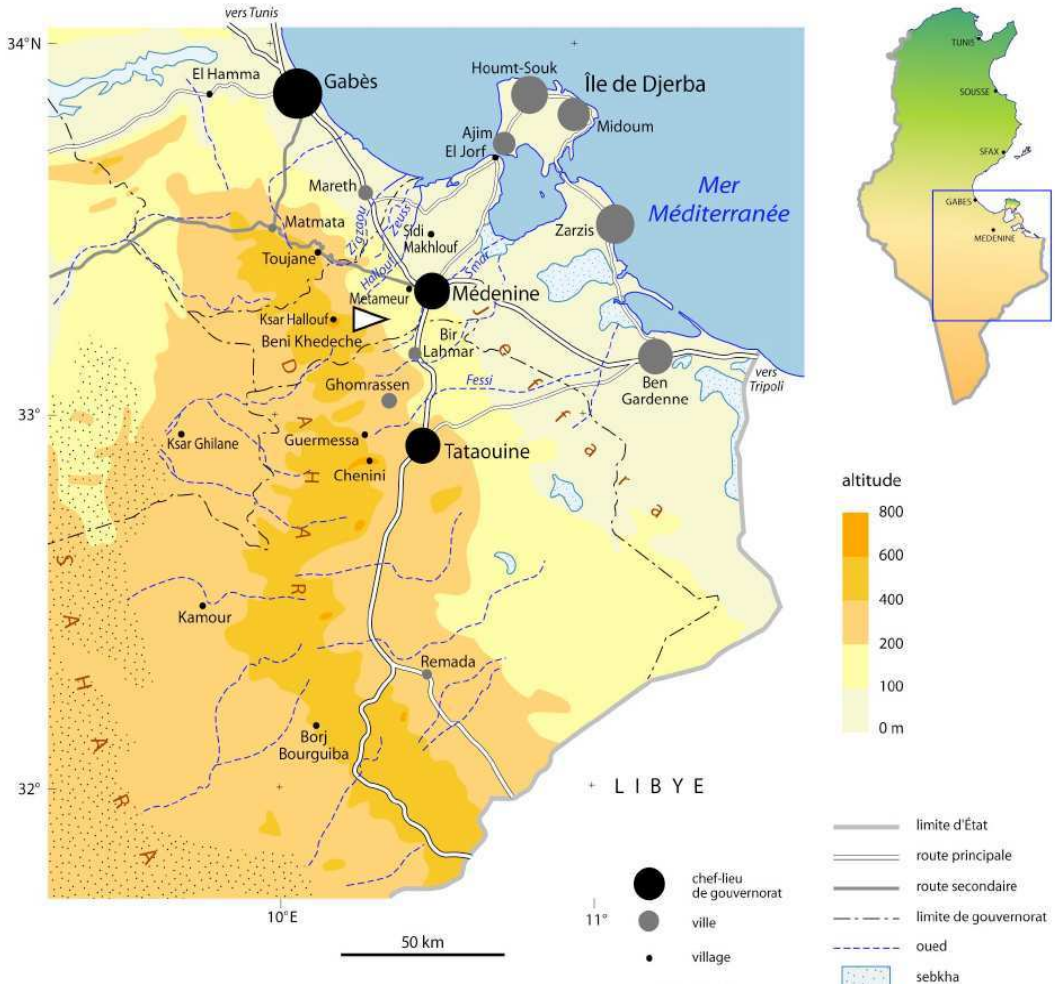

Figure 2. Study area: Zeuss-Koutine region

The groundwater resources are scarce and over exploited. This exploitation reaches 183% with annual renewable resource of 1.39 Mm^3. Two subsystems can be distinguished: the sub-system of private irrigated farms is based on surface wells (655 farms). The subsystem of public irrigation schemes is based on collective tube-wells (158 farms), normally established by the state. The water management is ensured by a water user association known as the 'GDA'. The agricultural production is based on crop production.

The data for this study were taken from a different surveys conducted in the laboratory of Economics and Rural Society between April 2005 and August 2009. A sample of 100 framers from these surveys was used in this analysis. The sample of farms was taken from a specific region geographical region in Sidi Maklouf and Médenine Sud districts. These districts represent 85.5 % of the total irrigated land area in the governorate of Medenine and the water scarcity and the increasing pressure on these ground water resources calls for a more efficient.

Table 1 Distribution of private irrigated farms surveyed by delegation and by land area

Delegations		Private farms		
	<= 1Ha	1-2 ha	>2	Total
Sidi Maklouf	13	18	19	50
Médenine Sud	10	21	19	50
Total study area	23	39	38	100

The selected sample comprises 23 farms smaller than one hectares (23% of the sample), 39 ranging between one and two hectares (39%) and 38 larger than two hectares (38%). It represent 13% (213 ha) of the total private irrigated land area in Zeuss-Koutine region.

These surveys involved a personal interview and it collected quantities and costs of inputs used in production, quantities and value of output, the quantity of water consumed and irrigation practices.
The current study considered the three main crops produced on the farms: cereals, olive tree and legumes. The production data included the output levels of each of these three groups of crops converted into monetary values. Aggregated inputs considered in this analysis are: (1) land measured in hectares, (2) irrigation water measured in m^3, (3) total labor measured in working days, (4) chemicals inputs measured in Tunisian dinars (TND), (5) and others costs, comprising the rest of inputs used (mechanisation, etc). Summary statistics of these variables is given in table 2.

Table 2 Descriptive statistics of the variables used in the DEA analysis

	Output	Inputs				
	Production (in TND)	Land (ha)	Water (m^3)	Labor (days/year)	Chemical inputs (in TND)	Others costs (in TND)
Average	14933,4	2,13	23306,4	1128,89	2091	992
Standard deviation	12016,19	1,31	22998	573,08	2141,41	1266,55
Minimum	1000	0,5	1800	288	35	0
Maximum	60000	9	128304	2957	8999	8900

Note: 1 TND (Tunisian dinar)=0,60 euros

4. Results and discussion

4.1 Technical and irrigation water efficiency

The technical efficiency (equation 1) is estimated using the program DEAP (Coelli, 1996) and irrigation water efficiency (equation 2) were modelled in the General Algebraic Modelling System software (GAMS) using the methodology proposed by Speelman et al. (2007).
Results for estimates of technical efficiency (TEi) and irrigation efficiency (IEi) are presented in table 3. The first thing to note about these results is that under constant returns to scale (CRS), the efficiency score derived is either less or equal to the efficiency score derived for the variable returns to scale (VRS) specification for every farms. This reflects the fact that, under VRS, inefficient farms are only compared to efficient farms of a similar size. For this reason, more farms are efficient under the VRS formulation.

Under the VRS specification the estimated input-oriented technical efficiency ranges from a minimum of 27.5% to 100% with an average estimate of 80.3%. This results means that a 19.7% decrease in all inputs is possible with present state of technology and unchanged outputs, or the same level of output can be reached by only using 80.3% of the used inputs, if technical inefficiency is completely removed. Thus, improving technical efficiency will significantly increase farm's revenue and profit.
The decomposition of the technical efficiency measure produced estimates of 19.7 per cent pure technical efficiency inefficiency and 23.9 per cent scale inefficiency. By eliminating scale inefficiency the farms can increase their average technical efficiency level from 64 to 80.3 per cent.

On the other hand, mean water efficiency is found to 47.8% and 60% under CRS and VRS, formulation, respectively, witch is much lower than technical efficiency and also exhibits greater variability, ranging from 1.5% to 100%. Under VRS assumption, the estimated irrigation water efficiency implies that the farms should be able to produce the same level of output using only 60% of its current level of irrigation water, while keeping other inputs constant, or that observed level of irrigated production could have been maintained by using the observed values of other inputs wile using 40% less irrigation water. This means that farmers can achieve significant savings in water use by improving the way they use the irrigation system and by using more advanced irrigations techniques.

Table 3 Frequency distribution of efficiency scores for the studied farms sample

Efficiency (%)	Technical efficiency (TE)		Irrigation efficiency (IE)	
	CRS	VRS	CRS	VRS
	Number of farms	Number of farms	Number of farms	Number of farms
0<E<=25	9	0	35	26
25<E<=50	28	10	24	18
50<E<=75	27	30	15	15
75<E<=100	36	60	26	41
N	100		100	
Mean efficiency	64	80.3	47.8	60
Min. efficiency	16	27.5	0.3	1.5
Max. efficiency	1.00	1.00	1.00	1.00
Scale Efficiency	76.1		74.5	

Table 4 gives the TE and IE distribution of three groups of farms identified as follows- (i) the least efficient (i.e. technical efficiency between 0 and 50%), (ii) the next most efficient (technical efficiency between 50% and 75%), and (iii) the most efficient overall (technical efficiency between 75 and 100%).

Table 4 Average efficiencies of selected farms, given constants return to scale assumption

	Average technical efficiency	Average irrigation efficiency
Group 1 (0<TE<=50%)	35%	22,58%
Group 2 (50 %< TE<=75%)	62%	60,7%
Group 3 (75%<TE<=1)	95%	90%

The first thing to note about these results is that under constant returns to scale (CRS), the average irrigation efficiency score derived is either less to the average technical efficiency score derived for every groups. In fact, while the average technical efficiency is approximately 95% for the third group, the mean IE of this group is only about 90%. For the first group, the average TE is around 35%; however, IE, for this group is only 22, 58%

Figure 3 give a graphical representation of the cumulative efficiency distributions for the different measures. Again it is clear that under both returns to scale specifications more farms were highly inefficient in the use of water compared to overall technical efficiency.

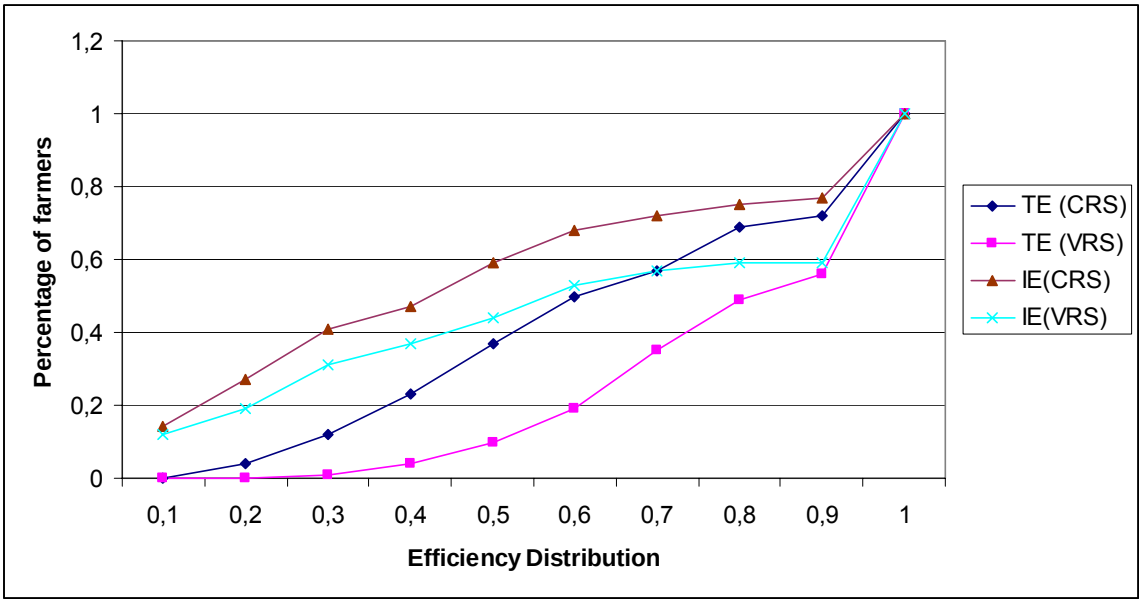

Figure 3. Cumulative efficiency distribution for both technical and irrigation efficiency under VRS and CRS specification

We will now consider the annual mean water excess across all groups of farms as shown in table 5. The average water excess range from 0 m^3/ ha to 24570.89 m^3/ ha, with an average estimate of

2421.61 m^3/ha under CRS assumption. Considering that the mean irrigation water use per farm was 4652.90 m^3/ha in the study area, this infers that almost half of the water use was “excess”.

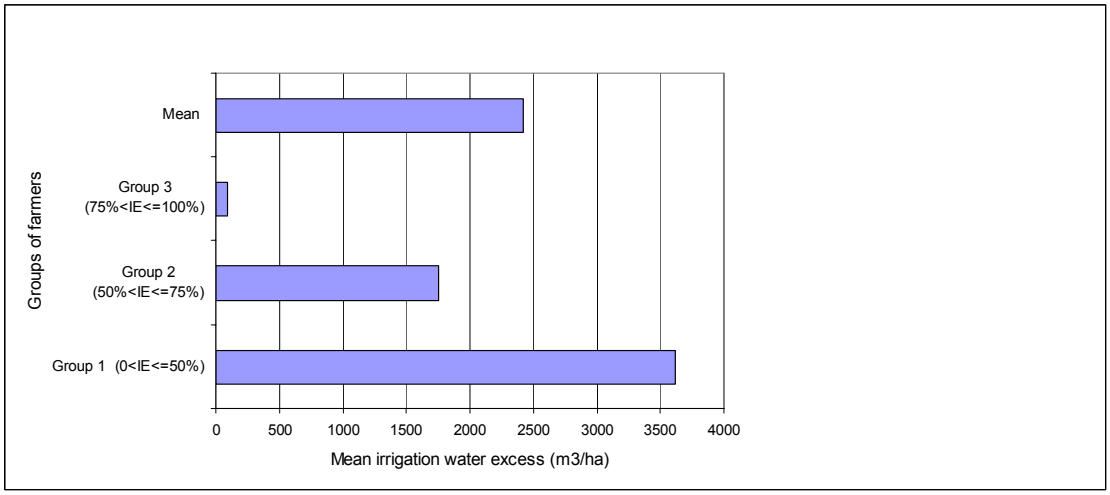

Figure 4. Estimated irrigation water excess per hectare (m3/ha) by groups of farmers, given constants return to scale assumption

The Pearson correlations between TE and IE measures are presented in table 5. These estimates provide evidence that there is a weak positive correlation between technical efficiency and irrigation efficiency under CRS and VRS specification.

Table 5 Pearson correlations between efficiency measures

	TE (CRS)	TE (VRS)	IE (CRS)	IE (VRS)
CRS	1			
VRS	0,67**	1		
Sub CRS	0,85**	0,69**	1	
Sub VRS	0,609**	0,875**	0,79**	1

Note: ** indicates a 99% significance level

A paired sample t-test to analyse the equality between IE efficiencies and TE was statistically significant. Furthermore, irrigation efficiency was significantly lower then technical efficiency measures, both under CRS and VRS specification (table 6).

Table 6 Paired samples t-tests demonstrating the difference between technical efficiency and irrigation efficiency

	Mean difference	Std dev.	t-statistic
CRS: IE-TE	-0,21	0,21	-7,24**
VRS: IE-TE	-0,16	0,19	-6,21**

Note: ** indicates a 99% significance level

Two conclusions arise from our analysis of inefficiency. First, the farmers and agencies that are involved in agricultural development programs need to appreciate there is a technical inefficiency problem. Operating at best practice, farm families would be able to release 36% of all inputs for use in alternative economic activities to generate extra income for family’s welfare. Surplus resources such as water could be reallocated to other water demands.

5. Discussion and summary

Sub-vector Data Envelopment analysis has been applied here for the first time to measures irrigation water use efficiency and to quantify the reduction potential, or excess, of irrigation water used at the farm level.

Many reforms have been made in Tunisia in the last two decades to improve irrigation water use efficiency. The research shows that the results of these reforms are not always guaranteed. In Zeuss-Koutine region, characterized by small scale agricultural systems and government subsidies for water

saving technologies, the calculated irrigation water use efficiency is still low and does not reflect the water-saving orientated policies that have been applied. The mean IE is 47.8% under CRS assumption which is very low, particularly for a semi-arid country, such as Tunisia, with limited water resources. This implies that there exists a potential 52.2% reduction in water use if all farms operated efficiently. Considering that the mean irrigation water use per farm was 4652.90 m^3/ha in the study area, this infers that almost half of water used was "excess". These low efficiencies suggest that substantial decreases in water use can be attained given existing technology, without compromising the key role in rural development played by small-scale irrigation. Substantial water inefficiencies in Tunisia (47%) were reported also by Dhehibi et al. (2007) for irrigated citrus production in Cap Bon (this region is the primary consumer of irrigation water in Tunisia). Albouchi et al. (2005) found that IE in the Kairouan region (the main agricultural area in the central part of Tunisia) was approximately 53%. Also, Frija et al. (2009), found that IE of small-scale greenhouse farmers in the Teboulba region was approximately 41, 8% under constant returns to scale specification (CRS). Thus, Tunisia still has much to do to improve the use and sustainability of its water resources, starting at the farm level.

In order to enrich this analysis, the source of efficiency differentials between farmers needs to be explained by further research of socio-economic characteristics. This topic has been addressed by other researchers and a common finding is that increase age and educational levels are positively related to technical efficiency (Dhibi et al, 2007; Frija et al, 2009). However, it is worth noting that the present study has some limitations. Because of the need to rely on farmers memories, the efficiency analysis is based generally on a single season. Extrapolating the results to other areas, years and seasons needs to be done with care. Furthermore, factors such as the timing of fertiliser application and irrigation can have an impact on efficiency. Therefore inefficiency is not just a result of the differences in the quantities of land, human labour, seed and mechanical labour, irrigation and chemicals revealed in this research. Also, institutional factors such as extension, systems research and general policies need to be examined. Future research needs to concentrate on developing the appropriate frontier model that encompasses all components of the whole farming systems quality and time variation components. Explicitly incorporating these other factors in the analysis can only give rise to improvements in measured efficiency levels.

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